

MANGENESIS — Hackathon Presentation Guide & 6-Student Pitch Strategy

Problem Statement SIH26009 | Ministry of Steel — MOIL Ltd. | AI Space Intelligence for Manganese Reserves

1. Executive Overview & Problem Definition

MANGENESIS is an AI-powered space intelligence platform engineered for **MOIL Ltd.** to simultaneously solve two major industry challenges:

- **Challenge 1: Slow & Expensive Ore Exploration:** Traditional mineral prospecting relies on blind field surveys and exploratory drill holes costing crores. MANGENESIS uses 6 multi-spectral satellite streams and XGBoost AI to locate manganese deposit zones with **88.25% ROC-AUC accuracy** before breaking ground.
- **Challenge 2: Surprise Production Shortfalls:** Flooded haul roads, machine hydraulic degradation, and blasting delays lead to sudden production deficits. MANGENESIS predicts output deficits **7 days in advance** using LightGBM, isolates root causes with TreeSHAP, and optimizes fleet recovery with MILP.

2. Technical Terminology in Simple Words (Cheatsheet)

Term	What It Means in Simple Plain English	Project Metric / Context
XGBoost	A machine learning classifier that learns underground manganese signatures from multi-spectral satellite imagery and historical drill assays.	0.8825 ROC-AUC target confidence
LightGBM	A fast gradient-boosted time-series regressor forecasting daily production 7 days ahead based on weather, shifts, and equipment telemetry.	Predicts output up to 7 days ahead (94% conf)
TreeSHAP	Game-theoretic Explainable AI (XAI) that decomposes exactly <i>WHY</i> production drops (e.g., 58% Equipment + 24% Rain + 18% Blasting).	Eliminates AI black-box for mine managers
MILP	Mixed Integer Linear Programming: An optimization math solver that calculates the highest-yield tactical actions (e.g., redeploying standby loaders).	Recovers up to 77% of projected shortfall tonnage
Sentinel-2	European Space Agency (ESA) multi-spectral satellite capturing 13 spectral bands including SWIR to map manganese alteration.	10m resolution, 5-day revisit cycle (100% Free)
NASA GPM & SMAP	NASA radar satellites measuring precipitation and ground soil moisture to detect haul road waterlogging and pit inundation.	Daily rainfall (mm) & soil moisture (%) live ingestion
PostGIS + Supabase	Cloud PostgreSQL spatial database holding polygonal zone boundaries, diamond core drill pins, and real-time alert logs.	Enterprise RLS security & WebSockets real-time sync

3. The Core Three-Step Innovation: Predict → Explain → Act

Stage 1: PREDICT (7 Days Ahead)	Stage 2: EXPLAIN (Root Causes)	Stage 3: ACT (MILP Optimizer)
LightGBM Regressor Forecasts daily output vs 10,000 TPD target. Flags Day 4–6 high-risk deficit (e.g. -2,200 T shortfall projected).	TreeSHAP Feature Attribution Isolates exact root causes: 58% Primary Shovel Breakdown, 24% Road Mud Inundation, 18% Blasting Delay.	MILP Solver Execution Generates prioritized recovery actions: 1. Redeploy Standby Shovel (+1,100T), 2. Reroute Haul Ramp (+520T) → Restores 77% deficit.

4. 6-Student Presentation Division & Speaking Roles

Member & Role	Screen to Demo	Key Talking Points (30–45 Secs)	Judge Q&A; Defense
Student 1 Team Lead & Architect	Command Center (MOIL Mine Selector)	- Problem statement (SIH26009, MOIL). - Multi-mine switch across 6 sites. - Live KPI cards (Output, Target, Risk %).	<i>Q: Scalability?</i> A: Works for all 6 MOIL mines & any global mine with zero code changes.
Student 2 Satellite GIS Engineer	Reserve Intelligence (Data Fusion Panel)	- Ingestion of 6 free NASA/ESA satellites. - Multi-band stacking (Sentinel-2 NDVI, GPM rain, SMAP moisture, MODIS thermal).	<i>Q: Satellite data cost?</i> A: 100% Free & open access from Copernicus & NASA Earthdata.
Student 3 AI Exploration Lead	Reserve Intelligence (Leaflet GIS Map)	- XGBoost model trained on historical assays. 0.8825 ROC-AUC target identification. - Inspect Mn grade & recoverable tonnes.	<i>Q: How to detect Mn from orbit?</i> A: Hydrothermal alteration exhibits unique SWIR spectral absorption bands.
Student 4 Production Forecaster	Production Forecast (14-Day & 7-Day Chart)	14-day historical actuals → 7-day AI forecast. - Early warning 7 days before shortfall hits feed. - Low, Medium, High risk period windows.	<i>Q: Why LightGBM over LSTM?</i> A: 10x faster training on tabular telemetry, robust to missing rows, native TreeSHAP.
Student 5 Explainable AI (XAI)	Risk Analysis (Root Cause Donut)	- TreeSHAP game-theoretic feature attribution. - Decomposes risk into exact percentages. - Live telemetry signal monitoring stream.	<i>Q: Why is TreeSHAP needed?</i> A: Mine managers need mathematical accountability, not black-box predictions.
Student 6 Optimization & Alerts	Action Center & Alert Center (Virtual Phone Demo)	- MILP solves optimal shovel redeployment. - Restores 77% deficit tonnage. - Live SMS trigger on virtual mobile simulator + desktop push chime.	<i>Q: How are actions chosen?</i> A: MILP optimizes Impact Score constrained by shovel fleet & haul cycle times.

5. Complete 2-Minute Master Presentation Pitch Script

[0:00 – Student 1]: “Respected judges, we are Team MANGENESIS. MOIL produces over 45% of India’s manganese, but faces two crippling challenges: exploratory drilling takes months and costs crores, while surprise production shortfalls cost lakhs daily. MANGENESIS solves both using space intelligence.”

[0:30 – Student 2 & 3]: “We ingest 6 free NASA and ESA satellite streams — including Sentinel-2 multispectral imagery and NASA soil moisture. Our XGBoost model maps manganese alteration zones underground with 88.25% ROC-AUC accuracy, saving millions in exploratory drilling.”

[1:00 – Student 4 & 5]: “For operational continuity, our LightGBM model predicts production shortfalls 7 days in advance. TreeSHAP explains the exact root causes — like 58% excavator pressure drop and 24% haul road mud saturation — eliminating the AI black box.”

[1:30 – Student 6]: “Finally, our MILP optimizer computes the highest-yield recovery actions to restore up to 77% of lost tonnage, while our multi-channel escalation system dispatches instant SMS, email, and desktop push alerts to mine controllers. We are now ready for your questions.”